

Spatial Data Analysis

AGRON 5130 - Module 12 Assignment

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Introduction

The goal of this assignment is to practice the full workflow from Module 12:

- loading and inspecting spatial vector data
- checking and using coordinate reference systems
- joining spatial datasets
- interpreting scale and resolution effects
- building interpolated surfaces (rasters) with kriging
- converting predictions into management recommendations

You will complete two parts:

- **Part 1:** vector data exploration and spatial joins
- **Part 2:** raster creation and spatial pattern analysis

Submit a knitted HTML/PDF generated from your R Markdown file to Canvas.

Setup

```
library(sf)
```

```
## Linking to GEOS 3.12.1, GDAL 3.8.4, PROJ 9.4.0; sf_use_s2() is TRUE
```

```
library(gstat)  
library(stars)
```

```
## Loading required package: abind
```

Part 1: Vector Data and Spatial Joins

Question 1

2 points

Read in the SSURGO polygon layer (`data/sample-ssurgo.shp`) and the soil test data (`data/soil_tests.shp`).

```
ssurgo <- read_sf('./data/sample-ssurgo.shp')
soil <- read_sf('./data/soil_tests.shp')
```

For each dataset, report:

1. the geometry type
2. the CRS
3. whether the CRS is longlat or projected

Interpretation: If the CRS is longlat, which one should you transform to a projected system, and why?

Question 2

2 points

Transform both datasets to the same projected CRS (use 32615 for UTM zone 15N).

```
ssurgo <- st_transform(ssurgo, 32615)
soil <- st_transform(soil, 32615)
```

Make a map showing both layers together. Use `plot()` to overlay SSURGO polygons (border only) with soil test points on top.

Question 3

2 points

Use `st_join()` with `st_within` to assign SSURGO attributes (for example `mukey`, `muname`, `om_r`) to each soil test point.

```
soil_b <- st_join(soil, ssurgo, join = st_within)
soil_b
```

```
## Simple feature collection with 700 features and 24 fields
## Geometry type: POINT
## Dimension: XY
## Bounding box: xmin: 522632.5 ymin: 4872027 xmax: 522944.7 ymax: 4872720
## Projected CRS: WGS 84 / UTM zone 15N
## # A tibble: 700 x 25
##      obs Sample_id SampleDate ReportDate    P2 Grower   Field attribute measure
## * <int>      <int> <chr>         <chr>    <int> <chr>   <chr> <chr>          <dbl>
## 1     1         1    10/26/2018 10/30/2018     0 Tom Besch Paul~ Om         4.2
## 2     2         2    10/26/2018 10/30/2018     0 Tom Besch Paul~ Om         3.1
## 3     3         3    10/26/2018 10/30/2018     0 Tom Besch Paul~ Om         3.6
## 4     4         4    10/26/2018 10/30/2018     0 Tom Besch Paul~ Om         3.9
## 5     5         5    10/26/2018 10/30/2018     0 Tom Besch Paul~ Om         4.7
## 6     6         6    10/26/2018 10/30/2018     0 Tom Besch Paul~ Om         3.6
## 7     7         7    10/26/2018 10/30/2018     0 Tom Besch Paul~ Om         4.9
## 8     8         8    10/26/2018 10/30/2018     0 Tom Besch Paul~ Om         4.1
## 9     9         9    10/26/2018 10/30/2018     0 Tom Besch Paul~ Om          5
## 10    10        10    10/26/2018 10/30/2018     0 Tom Besch Paul~ Om         4.1
```

```
## # i 690 more rows
## # i 16 more variables: geometry <POINT [m]>, mukey <int>, area_ac <dbl>,
## #   cokey <chr>, muname <chr>, compct <int>, hzdept <int>, om_r <dbl>,
## #   clyttl_ <dbl>, sltttl_ <dbl>, sndttl_ <dbl>, cec7_r <dbl>, ph1t12_ <dbl>,
## #   awc_r <dbl>, ksat_r <dbl>, dbthrd_ <dbl>
```

Question 4

2 points

Compute the mean measure value by SSURGO map unit and attribute (use `aggregate()`).

Include `mukey` in the grouping variables.

```
aggregate(measure ~ mukey,
          soil_b,
          FUN = mean)
```

```
##      mukey  measure
## 1  395372  130.6011
## 2   784798  118.6120
## 3 1384967  156.5700
```

Part 2: Raster Creation and Spatial Pattern

Question 5

1 point

Load the field boundary (`data/boundary.shp`) and reproject (using `st_transform`) it to EPSG 32615. Create a raster grid that covers the boundary and has a 10 m resolution using `st_as_stars()` and `st_crop()`.

Plot your empty grid.

```
boundary <- st_read('./data/boundary.shp')
```

```
## Reading layer 'boundary' from data source
##   '/home/caio/Documents/module-10/module-12/assignment/data/boundary.shp'
##   using driver 'ESRI Shapefile'
## Simple feature collection with 1 feature and 1 field
## Geometry type: POLYGON
## Dimension:      XY
## Bounding box:   xmin: 522575.4 ymin: 4871972 xmax: 522982.6 ymax: 4872753
## Projected CRS: WGS 84 / UTM zone 15N
```

```
boundary <- st_transform(boundary, 32615)
```

```
grd <- st_as_stars(st_bbox(boundary),
                  dx = 10,
                  dy = 10)
```

```
grd <- st_crop(grd, boundary)
plot(grd)
```

```
## Warning in plot.stars(grd): breaks="quantile" leads to a single class; maybe
## try breaks="equal" instead?
```



Question 6

2 points

Choose the soil attribute `P_bray` from the original soil test points (not the boundary). Compute the empirical variogram with `variogram()` and plot it.

```
soil$attribute |> unique()
```

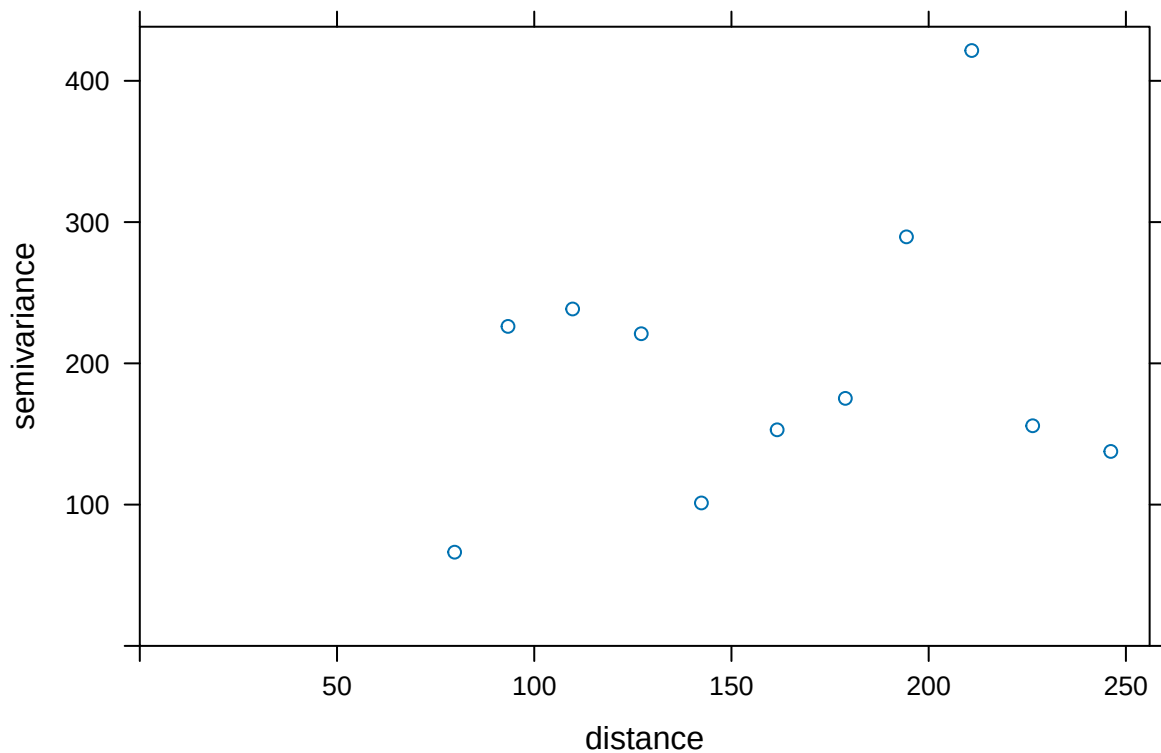
```
## [1] "0m"      "Ph"      "Bph"     "No3"     "Salinity" "P"
## [7] "P_bray"  "P_olsen" "M3_p"    "M3icp_p" "K"        "Ca"
## [13] "Mg"      "Na"      "S"       "Zn"      "Cu"       "Mn"
## [19] "Fe"      "B"       "Cec"     "Ca_sat"  "Mg_sat"   "K_sat"
## [25] "H_sat"
```

```
# Example: select pH
selected <- soil[soil$attribute == "P_bray", ]
v <- variogram(measure ~ 1, selected)
v
```

```
##      np      dist      gamma dir.hor dir.ver      id
```

```
## 1  3  79.81215  66.33333      0      0 var1
## 2 26  93.34996 226.17308      0      0 var1
## 3  9 109.72972 238.50000      0      0 var1
## 4 15 127.13151 220.96667      0      0 var1
## 5 14 142.38805 101.17857      0      0 var1
## 6 11 161.58679 152.95455      0      0 var1
## 7  7 178.87073 175.21429      0      0 var1
## 8 16 194.37063 289.56250      0      0 var1
## 9 23 210.91696 421.45652      0      0 var1
## 10 22 226.36641 155.81818      0      0 var1
## 11 13 246.17398 137.65385      0      0 var1
```

```
plot(v)
```



Question 7

2 points

Using your selected attribute from Question 6, fit a spherical model ("Sph") with `fit.variogram()` and plot empirical + fitted variogram together.

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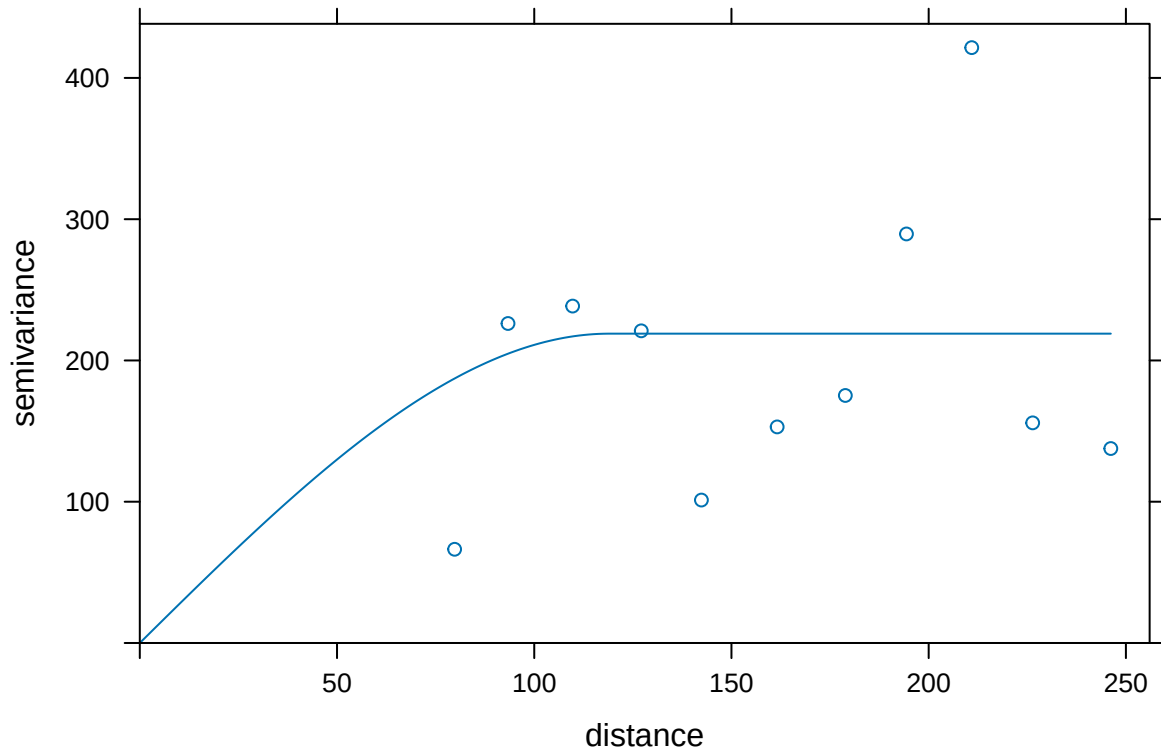
```
m <- fit.variogram(v, vgm("Sph"))
```

```
## Warning in fit.variogram(v, vgm("Sph")): singular model in variogram fit
```

```
m
```

```
##   model  psill  range
## 1   Nug    0.000  0.000
## 2   Sph 218.977 119.031
```

```
plot(v, model = m)
```



Question 8

2 points

Using the fitted model from Question 7, run `krige()` across the grid and map the predicted surface (`var1.pred`).

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```
kr <- krige(measure ~ 1, selected, grd, model = m)
```

```
## [using ordinary kriging]
```

```
plot(kr["var1.pred"])
plot(st_geometry(boundary), add = TRUE, border = "black", col = NA, lwd = 1)
plot(st_geometry(selected), add = TRUE, pch = 16, cex = 0.5, col = "grey20")
```

